AI Service Orchestrator for Informal Economy

Project Overview

An AI-powered service coordination platform designed for Pakistan’s informal economy. The system helps users find nearby service providers, automate booking workflows, coordinate workers, and improve communication between customers and shops using AI-driven reasoning and automation.

Core Features

1. AI Intent Understanding

What it does

The AI understands user requests in English, Urdu, and Roman Urdu.

Example

"Mujhe kal AC technician chahiye"

AI Extracts

- Service Type
- Location
- Preferred Time

Implementation

- Gemini/OpenAI API
- NLP processing
- Structured extraction pipeline
- Antigravity workflow orchestration

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2. Exact Location Collection

What it does

Instead of only using area names like DHA, the system asks for:

- House address
- Map pin

- Temporary or permanent location

Why it matters

Improves technician navigation and reduces confusion.

Implementation

- Google Maps integration
- Flutter map picker
- Saved locations database

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3. Dual Rating System

What it does

Both customers and service providers can rate each other.

Benefits

- Builds trust
- Prevents fake requests
- Improves service quality

Implementation

- Rating database
- User reputation scoring
- AI ranking system uses ratings during provider selection

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4. Provider Matching & Ranking

What it does

AI selects the best provider based on:

- Distance
- Availability
- Ratings
- Worker specialization

Example

"Selected because nearest available inverter AC specialist"

Implementation

- Ranking algorithm
- AI reasoning engine
- Mock provider dataset or Maps API

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5. Service Provider Dashboard

What it does

Service providers can:

- View customer requests
- See issue details
- View location and timing
- Track booking status

Benefits

Improves communication and preparation before arrival.

Implementation

- Flutter dashboard UI
- Backend booking management system
- Firebase/Supabase database

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6. Worker Specialization System

What it does

Shopkeepers can register workers with different skills.

Example

Ali:

- AC installation
- Inverter repair

Usman:

- Gas refill
- Wiring

Benefits

AI can assign the right worker automatically.

Implementation

- Worker profile database
- Specialty tags
- AI assignment logic

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7. AI Availability Negotiation

What it does

The AI can negotiate appointment timings with the shopkeeper.

Example

AI: "Kal 10 baje possible hai?"

Shopkeeper: "11 baje bhej deta hoon"

Benefits

Makes scheduling more realistic and flexible.

Implementation

- Conversational AI workflow
- Simulated negotiation logic
- Antigravity orchestration

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8. Urdu AI Calling System

What it does

AI simulates calling nearby shops using Urdu language to confirm:

- Availability
- Timing
- Worker assignment

Benefits

Matches Pakistan's real service workflow.

Implementation

- AI conversation agent
- Urdu Text-to-Speech
- Speech-to-Text
- Simulated call UI
- Live transcript system

Recommended Approach

Use simulated AI calling instead of real telecom calls for hackathon stability.

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9. Live Service Tracking

What it does

Customers can track service status.

Example Status

- Worker Assigned
- Technician Left Shop
- Arriving in 15 Minutes
- Service Completed

Benefits

Improves transparency and customer trust.

Implementation

- Real-time booking updates
- Simulated tracking timeline
- Push notifications

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10. Chat System for Material Coordination

What it does

Customers and providers can chat before the visit.

Purpose

Allows workers to know required materials before arriving.

Benefits

Reduces repeat visits and saves time.

Example

"Please arrange outdoor unit access before arrival"

Implementation

- Real-time chat system
- Image sharing support
- AI issue summarization

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11. AI Material Suggestion

What it does

AI predicts possible materials/tools needed based on the issue.

Example

- Cooling gas
- Capacitor
- Pipe insulation

Benefits

Improves technician preparation.

Implementation

- AI issue analysis
- Knowledge-based recommendation engine

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12. Referral System Between Shops

What it does

If a shop is unavailable, it can refer another provider.

Benefits

- Better customer experience
- Higher booking completion rate
- Stronger service network

Implementation

- Provider network system
- Referral workflow engine
- AI rerouting logic

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13. Booking Simulation

What it does

The system simulates:

- Booking confirmation
- Technician assignment
- Scheduling
- Notifications

Why it matters

This is a mandatory hackathon requirement.

Implementation

- Mock APIs
- Database updates
- Simulated notifications
- Booking receipts

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14. Follow-Up Automation

What it does

The AI handles reminders and post-service communication.

Example

- Reminder before appointment
- Service completion confirmation
- Customer feedback request

Implementation

- Notification scheduler
- Automated workflow triggers
- AI-generated follow-up messages

Recommended Tech Stack

Layer	Technology
Mobile App	Flutter
Backend	FastAPI
AI Models	Gemini / OpenAI
Database	Firebase / Supabase
Maps	Google Maps API
Voice AI	Whisper + TTS APIs
Workflow Engine	Google Antigravity

Recommended Agent Architecture

1. Intent Agent

Understands user requests.

2. Location Agent

Handles address and map pin processing.

3. Provider Search Agent

Finds nearby service providers.

4. Ranking Agent

Chooses the best provider.

5. Negotiation Agent

Handles availability discussion.

6. Booking Agent

Creates booking and scheduling.

7. Follow-Up Agent

Handles reminders and updates.

8. Referral Agent

Redirects requests to alternative shops.

Project Strength

This project focuses on solving real-world problems in Pakistan's informal economy through AI-driven automation, coordination, and communication.

The system goes beyond a simple booking app by demonstrating:

- AI reasoning
- autonomous workflows
- provider coordination
- intelligent decision-making
- conversational automation
- workflow execution
- follow-up management

This strongly aligns with the hackathon's agentic AI requirements.